

Permissions

Permissions

- Who can access your files?
 - Owner (you)
 - Group
 - Others (everyone)
- What can they do?
 - Read
 - Write
 - Execute
- How many yes no options do we have?

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 - 9

Viewing permissions

- How do we see what files we have?
 - `ls`
- `ls` has an extra flag to tell us more about files:
 - `ls -l`

Breaking it down

```
langanth@LANGANTH-A63ED4:~$ ls -l
total 72
-rw-r--r-- 1 langanth langanth 27638 Sep  9 13:07 avengers.csv
-rw-r--r-- 1 langanth langanth 1080 Sep 11 14:24 capture.tar.gz
drwxr-xr-x 9 langanth langanth 4096 Sep 23 17:15 cis_241
drwxr-xr-x 3 langanth langanth 4096 Sep 15 12:58 ctf
-rw-r--r-- 1 langanth langanth 2407 Sep 18 12:24 cut.txt
drwxr-xr-x 2 langanth langanth 4096 Sep  2 12:28 demo_dest
drwxr-xr-x 2 langanth langanth 4096 Sep  2 12:30 demo_start
drwxr-xr-x 3 langanth langanth 4096 Sep 25 12:35 git_practice_2
drwxr-xr-x 3 langanth langanth 4096 Sep  3 20:22 projects
drwxr-xr-x 2 langanth langanth 4096 Sep 23 12:46 qq_testing
-rw-r--r-- 1 langanth langanth 0 Sep  9 17:03 script_practice.txt
drwxr-xr-x 2 langanth langanth 4096 Sep  9 12:33 scripting_fun
-rw-r--r-- 1 langanth langanth 420 Sep  9 17:03 typescript
```

```
drwxr-xr-x 3 langanth langanth 4096 Sep 15 12:58 ctf
```

Br

```
drwxr-xr-x 3
```

Number of links

What type of
file is this?
d = directory
l = link

Owner
permissions

Group
permissions

Other
permissions

```
langanth langanth 4096 Sep 15 12:58 ctf
```

Owner

Group

File size

Modified
date+time

Filename

Examples

`rwxrwxr-x`

`rw-r-----`

`-----rw-`

Changing Permissions

- new command: chmod (CHange MODe)
- Usage 1: chmod (who) (op) (perms) file
 - chmod u+r file
 - chmod o-rw file
 - Who: owner (u), group (g), other (o), all (a)
 - Perms: read (r), write (w), execute (x)

Changing Permissions

- Usage 2: `chmod ### file`
- Where each # is a number [0, 7]
- Take each possible permission:
 - 4 = read
 - 2 = write
 - 1 = execute
- and sum them up for owner group or other

Examples

- What is the number code for each of these?
 - `rw-rw-r-x`
 - `rw-r-----`
 - `-----rwx-`

Examples

- What is the number code for each of these?
 - rwxrwxr-x 775
 - rw-r-----
 - -----rW-

Examples

- What is the number code for each of these?
 - rwxrwxr-x 775
 - rw-r----- 640
 - -----rW-

Examples

- What is the number code for each of these?
 - rwxrwxr-x 775
 - rw-r----- 640
 - -----rw- 006